Al Balqa Applied University- Department of Computer Science		
Course Name	Theory of Computation and Automata	
Instructor	**Hasan Rashaideh**	
Assignment	**DFAs Design**	
Semester - Year	**2nd-2021-2022**	
Date	**Thursday, April 7, 2022**	

Problem 1. Design a corresponding DFA for the following languages:
(Note: use the JFLAP software tool to draw the answer and you can ignore the TRAP state)

1. $L = \{001,010,011\}$
2. $L = \{(aa)^n b : n \geq 1\}$
3. $L = \{w : w \in \{0,1\}^* \; and \; N_0(w) \; mod \; 3 = 1\}$
4. $L = \{ab^{4n}a : n \geq 0\}$
5. $L = \{w : w \in \{0,1\}^* \; and \; N_0(w) \leq 3 \; and \; N_1(w) \geq 0\}$
6. $L = \{waba : w \in \{a,b\}^*\}$
7. $L = \{babw : w \in \{a,b\}^*\}$
8. $L = \{w_1 bbaw_2 : w_1 \in \{a,b\}^* \text{ and } w_2 \in \{a,b\}^*\}$
9. $L = \{bw_1 bw_2 : w_1 \in \{a,b\}^* \text{ and } w_2 \in \{a,b\}^*\}$
10. $L = \{w : w \in \{a,b\}^* \text{ and } N_a(w) \; is \; even \; and \; N_b(w) \; is \; odd\}$